

Question 1: What is Power BI and why is it used in businesses?

Ans: **Power BI** is a business intelligence (BI) and data visualization tool developed by **Microsoft**. It helps organizations turn raw data into meaningful insights through interactive reports and dashboards.

Why businesses use Power BI:

1. Data Visualization

- Converts complex data into easy-to-understand charts, graphs, and dashboards

2. Data Integration

- Connects to multiple data sources (Excel, databases, cloud services, APIs)

3. Real-time Insights

- Allows businesses to monitor performance and make decisions quickly

4. Self-service Analytics

- Non-technical users can create reports without heavy IT support

5. Better Decision Making

- Helps identify trends, patterns, and opportunities

Question 2 : Name and explain the three main components of Power BI?

In Power BI, the three main components are:

1 Power BI Desktop

- A Windows application used to create reports and data models
- You can:
 - Import and transform data (Power Query)
 - Build relationships
 - Create visualizations and dashboards
- Mainly used by developers and analysts

2 Power BI Service

- **A cloud-based platform (web version)**
- Used to:
 - Publish reports from Desktop

- Share dashboards with others
 - Collaborate across teams
- Enables real-time access from anywhere

Power BI Mobile

- A mobile app for Android and iOS
- Allows users to:
 - View dashboards and reports on the go
 - Get alerts and insights anytime

Question 3 : Explain the Power BI workflow?

Ans: The Power BI workflow describes how data moves from raw sources to meaningful insights using Power BI.

Power BI Workflow (Step-by-step)

1 Data Collection (Get Data)

- Connect to various sources:
 - Excel, SQL databases, cloud services, APIs
- Data is imported into Power BI

2 Data Transformation (Power Query)

- Clean and shape the data:
 - Remove errors, filter rows, rename columns
 - Use transformations like Pivot / Unpivot
- Done in Power Query Editor

3 Data Modeling

- Create relationships between tables (Fact & Dimension)
- Define:
 - Primary/foreign keys
 - Measures using DAX (calculations)

4 Data Visualization (Report Building)

- Build interactive visuals:
 - Charts, tables, maps, KPIs
- Combine visuals into reports

5 Publishing

- Publish reports from Desktop to Power BI Service
- Makes reports available online

6 Sharing & Collaboration

- Share dashboards with users
- Set permissions and access control

7 Refresh & Monitoring

- Schedule automatic data refresh
- Monitor performance and update reports

Question 4 : List any four data cleaning tasks that can be performed in Power Query?

Ans: Data Cleaning Tasks:

1 Remove Duplicates

- Eliminates repeated rows to ensure data accuracy

2 Handle Missing Values

- Replace null values or remove rows with blanks

3 Change Data Types

- Convert columns to correct types (e.g., text → number, date)

4 Filter Rows

- Remove unwanted data (e.g., exclude nulls, filter by conditions)

Question 5 : Write step by step instructions to load the above dataset?

Ans: Here are step-by-step instructions to load a real estate dataset into Power BI

Steps to Load Dataset in Power BI

1 Open Power BI Desktop

- Launch Power BI Desktop on your system

2 Click on “Get Data”

- Go to the Home tab
- Click Get Data

3 Choose Data Source

- Select the source where your real estate dataset is stored:
 - CSV
- Click Connect

4 Select the File

- Browse and choose your real estate dataset file
- Click Open

5 Preview Data (Navigator Window)

- A preview window will appear
- Select the required table/sheet

6 Click “Load” or “Transform Data”

- Transform Data → open Power Query to clean data before loading

7 Verify Data in Data View

- Once loaded, check:
 - Columns and rows
 - Data types

8 Start Building Reports

- Use fields to create charts, maps, and dashboards

Question 6 : Define Data View, Report View, and Model View. Explain the purpose of each view

Ans: **1** Data View Purpose:

- Used to inspect and understand the data in tabular format

What you can do:

- View rows and columns of your dataset
- Check data types and values

- Create calculated columns

2 Report View Purpose:

- Used to create and design visual reports and dashboards

What you can do:

- Add charts, tables, maps, KPIs
- Apply filters and slicers
- Format visuals and layout

3 Model View Purpose:

- Used to **manage relationships between tables**

What you can do:

- Create/edit relationships (Fact & Dimension)
- View schema (star/snowflake)
- Set cardinality and filter direction

Question 7 : Discuss the different data sources that Power BI supports?

Ans: In Power BI, you can connect to a wide variety of data sources. These are generally grouped into categories:

1 File-Based Sources

- Excel files (.xlsx)
- CSV (Comma-separated values)
- Text files
- JSON, XML, PDF

Common for small to medium datasets

2 Database Sources

- SQL Server
- MySQL
- Oracle
- PostgreSQL

Used for structured, large-scale data storage

3 Cloud-Based Sources

- Azure services
- Google Analytics
- SharePoint Online
- Salesforce

Enables real-time and remote data access

4 Online Services & APIs

- Web APIs (REST services)
- OData feeds
- Social media and web data

Useful for dynamic and external data

5 Other Sources

- Power BI datasets
- R and Python scripts
- Dataflows

Advanced analytics and reuse of existing models

Question 8 : Split Owner Name to create two new columns as First Name and Last Name

Ans: before image

	A ^B _C Owner_Name	1 ² ₃ Contact
0%	Valid	90% Valid
0%	Error	0% Error
0%	Empty	10% Empty
2	Rahul Mehta	98
2	Anita Sharma	98
2	Ramesh Kumar	91
2	Neha Verma	99
3	Amit Joshi	90
2		93
2	Suresh Reddy	90
2	Divya Iyer	95
3	Manoj Patil	98

After picture

A ^B C First name			A ^B C Last Name			1 ² 3 Contact		
● Valid 90%			● Valid 90%			● Valid		
● Error 0%			● Error 0%			● Error		
● Empty 10%			● Empty 10%			● Empty		
2	Rahul		Mehta				98	
2	Anita		Sharma				98	
2	Ramesh		Kumar				91	
2	Neha		Verma				99	
3	Amit		Joshi				90	
2					null		93	
2	Suresh		Reddy				90	
2	Divya		Iyer				95	
3	Manoj		Patil				98	
2	Rai		Singh					

PROPERTIES

Name

Real_Estate_Data_100_Rows

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Split Column by Delimiter

Changed Type1

Renamed Columns